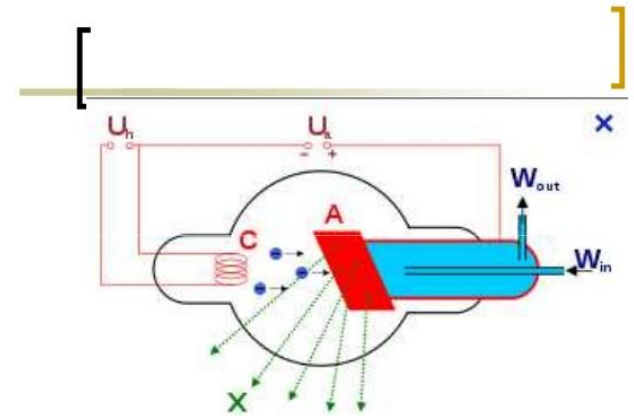




**11.4 Uses X – rays to
fulfill human needs.**

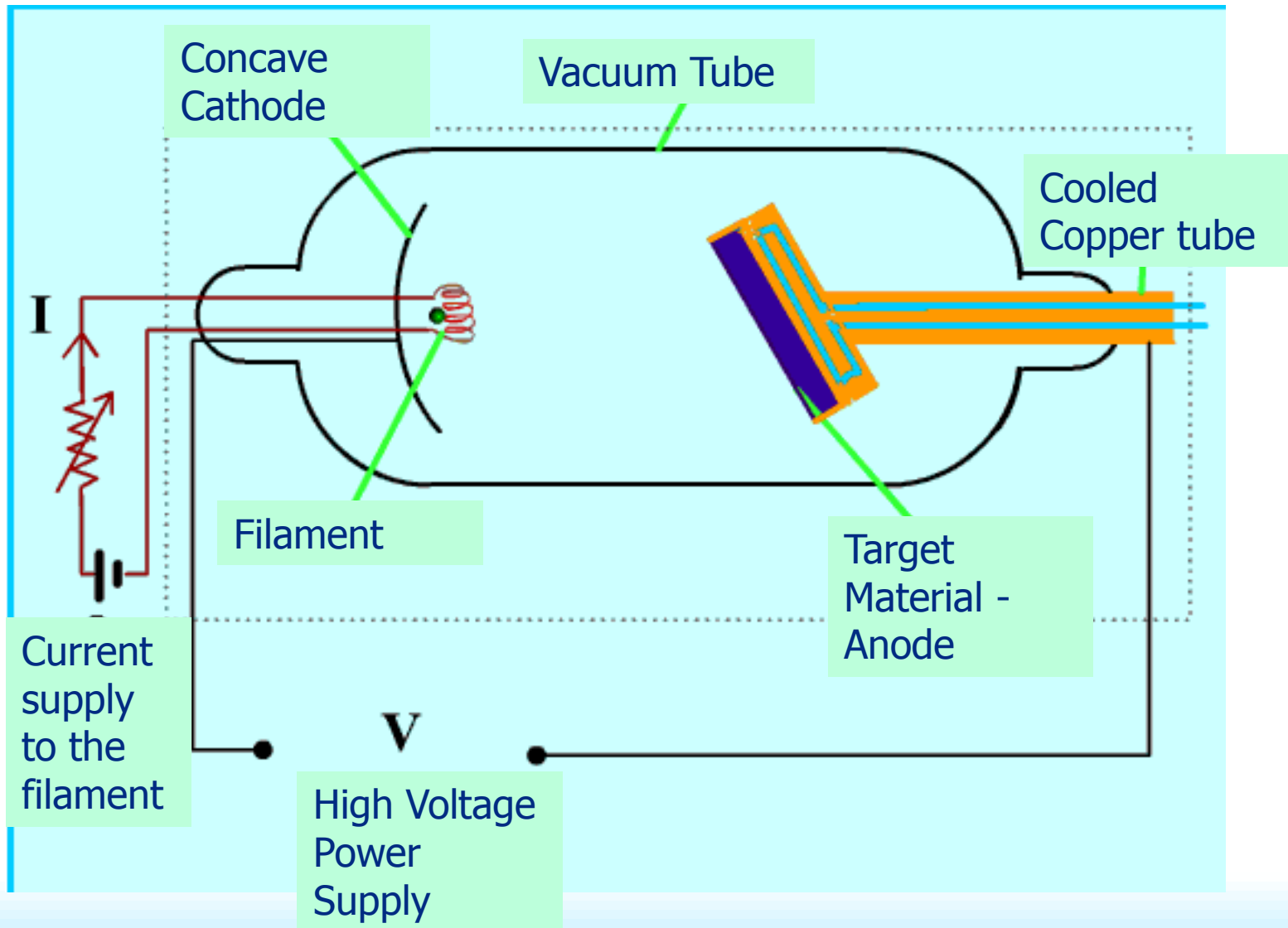
1. **Discovery of X-Ray**
2. **Production of X-Rays**
3. **Properties of X-Rays**
4. **Users of X-Rays in Different fields**

1. Discovery of X-Ray

- X- Ray was discovered by Wilhelm Röntgen in 1895.
- When he is doing a experiment with Cathode ray tube, he observed that some rays emit from the tube glowed a faint green on florescent sheet
- He also found that the new ray would pass through most substances casting shadows of solid objects on pieces of film.
- He named the new ray X-ray, because in mathematics "X" is used to indicated the unknown quantity.



1. Production of X-Rays



- tungsten(cathode) emits electron by thermionic emission in the vacuum tube at low pressure
- These electrons accelerated under a high potential difference and the electrons reach the metal with high kinetic energy.
- 99% of electron energy is converted to heat, so anode should be cooled by circulating water or oil around the anode.
- Metals with high melting points are used as targets, such as tungsten (W) or molybdenum (Mo).

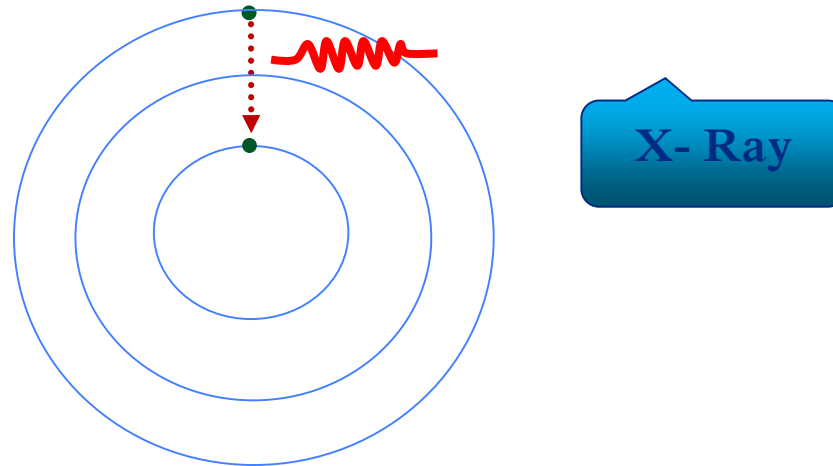
Two Types of X-Rays.

When the electrons collide with the target, about 1% of the resulting energy is emitted as X-ray. There are two types of X-rays.

1. Characteristics X-Rays

2. Bremsstrahlung X-Rays

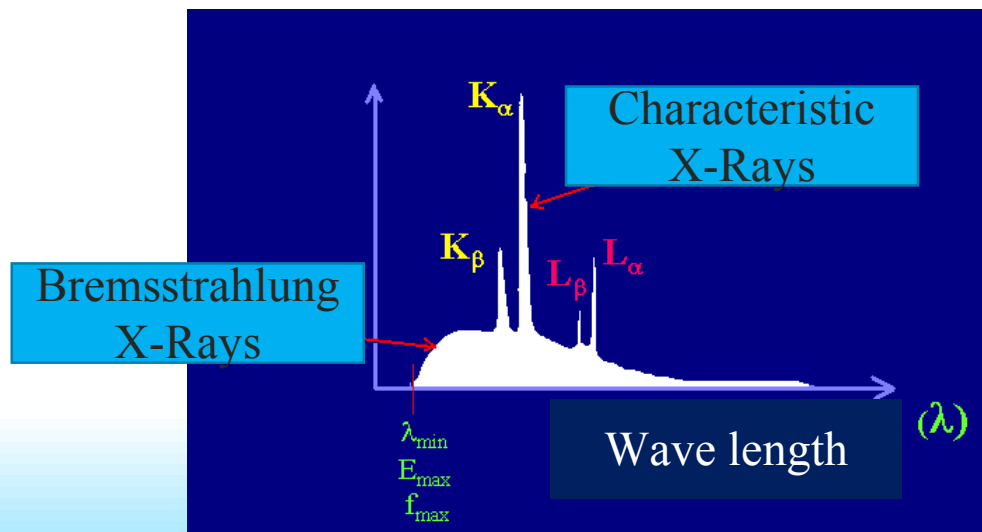
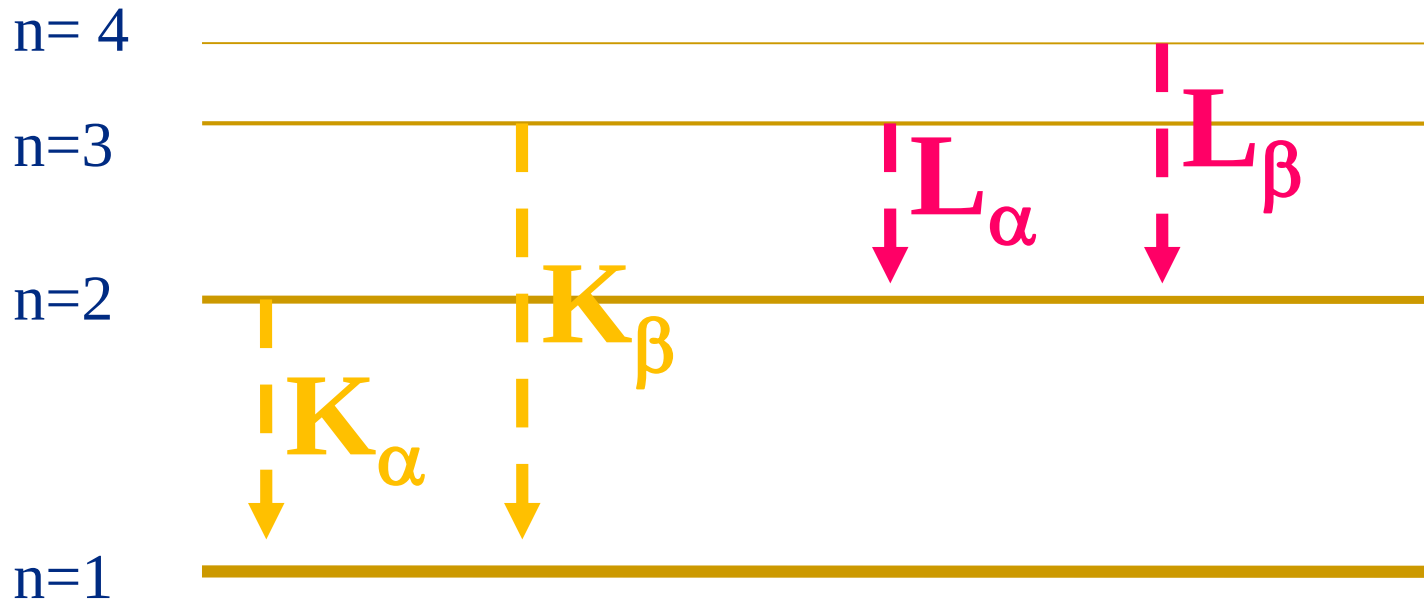
Characteristic_X-Rays



Kinetic Energy of the electron removes electron at lower energy level.

Then electrons in upper energy level jump to lower energy level by emitting a characteristic X-Ray

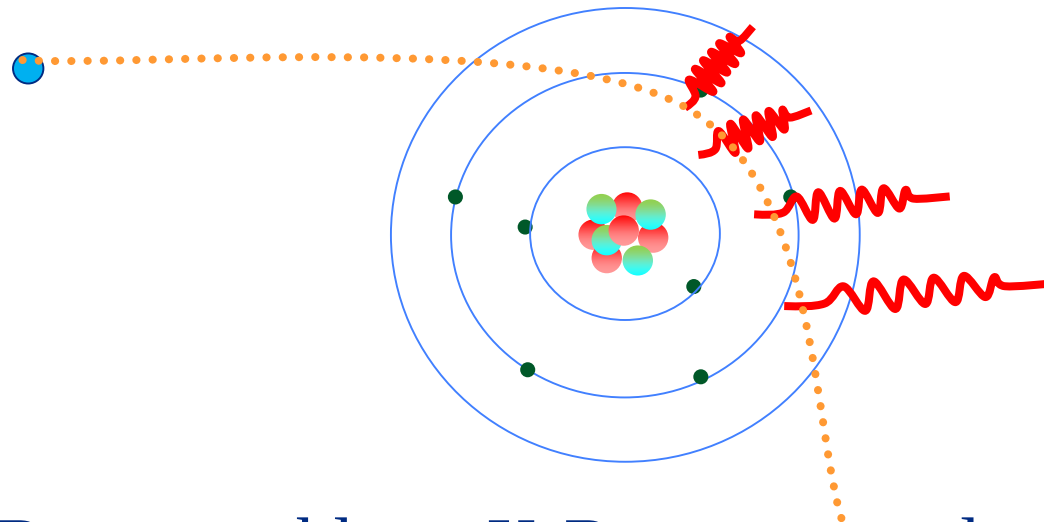
Spectrum of Characteristic X-Rays



Characteristics X-rays Line Spectrum

- This process produces an emission spectrum of X-rays at a few discrete frequencies, sometimes referred to as the spectral lines.
- The spectral lines generated depend on the target (anode) element used and thus are called characteristic lines. Usually these are transitions from upper shells into K shell (called K lines), into L shell (called L lines) and so on.
- Characteristic X-rays can be used to identify the particular element from which they are emitted.

Bremsstrahlung X-Rays



**Bremsstrahlung
Photons**

Bremsstrahlung X-Rays are produced by slowing down of the high Kinetic Energy electrons by the electric field surrounding the nuclei

Energy of the Bremsstrahlung X-Rays vary from zero to the maximum KE of electron

Maximum Energy of the Continuous Spectrum

- Kinetic Energy of a charge (charge of the electron- e) which is accelerated across a potential difference(V) $= eV$
- Energy of the produced X ray $= hf$
(h - Plank's constant / f – frequency)
- The electrons transfer all their energy into X-ray (photon energy).
- Maximum energy of this X ray is $E_{\max}$

$$E_{\max} = eV = hf_{\max}$$

$$f_{\max} = eV/h$$

$$V = f * \lambda$$

v - velocity of the wave

f - frequency

λ - wave length

$$\lambda_{\min} = hv/eV$$

- x-ray energies emitted by the system can be adjusted by changing the applied voltage.

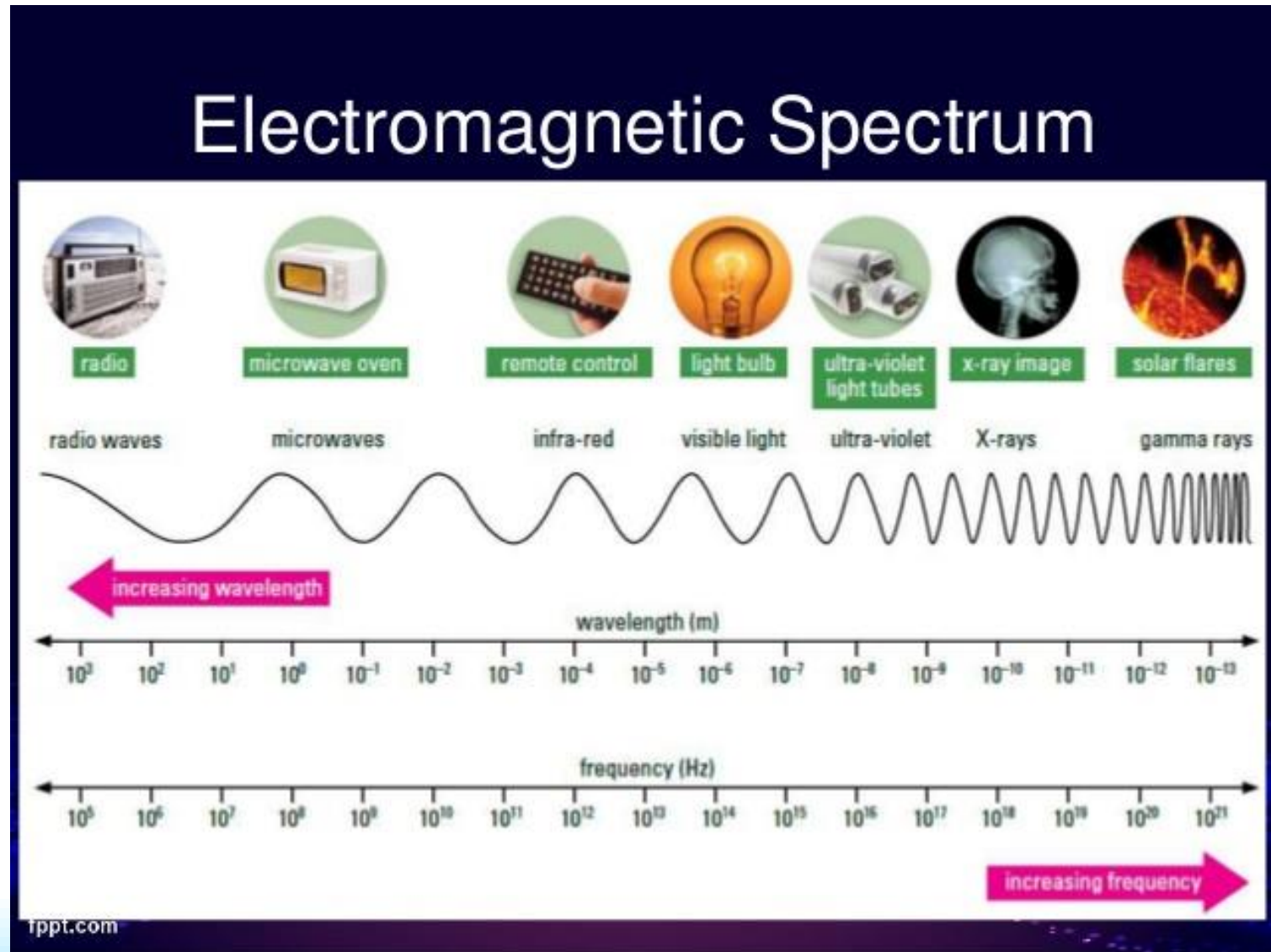
- X ray energy is dependent upon the potential difference between the anode and cathode.
- High energy (more penetration) X-ray can be produced with high voltage (V)

V $\propto$ Energy of the X ray

- The higher temperature of the filament which leads to the larger the number of electrons that leave the cathode and travel to the anode. The current (A) setting control the filament temperature, which relates to the intensity of the X-ray output.

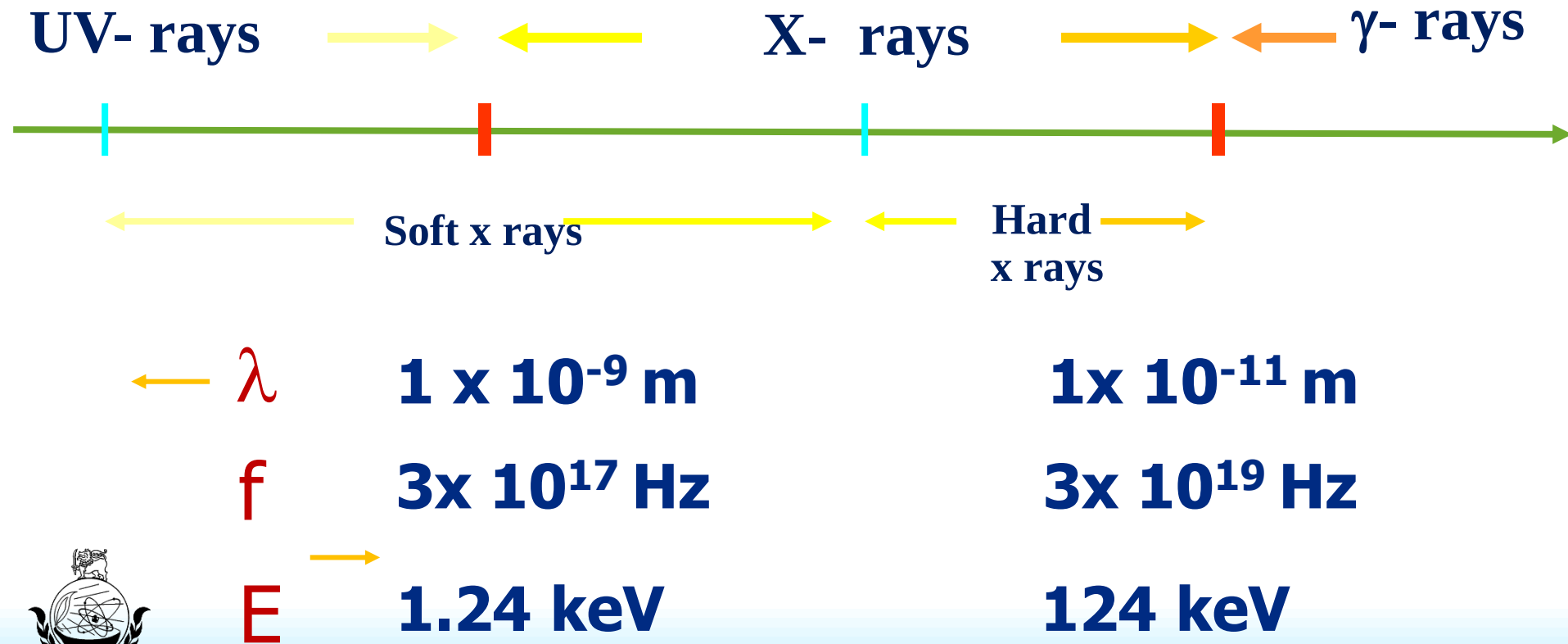
I $\propto$ Intensity of the X ray

3. Properties of the X ray



1. X-ray belong to a family of Electromagnetic radiation

- X ray having a wavelength (λ) between 1nm to 10pm.



2. Penetrating ability of the X ray

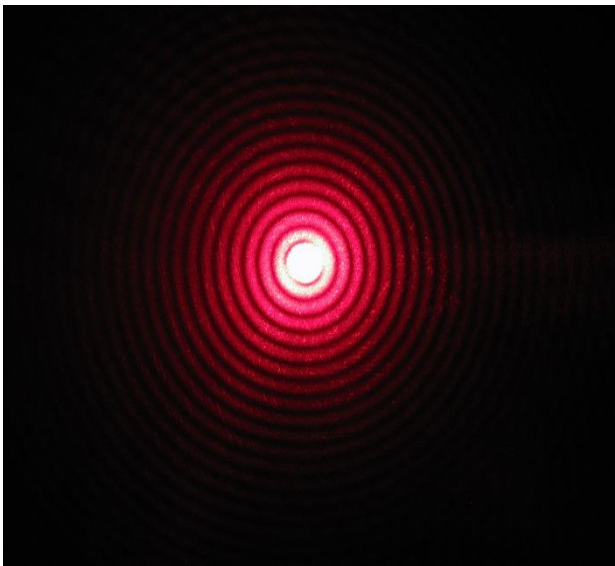
- Penetration ability depend only on the energy of the photon
 - X-rays with high photon energies (small wavelengths) are called hard X-rays – **higher penetration ability**
 - X-rays with lower photon energies (high wavelengths) are called soft X-rays – **lower penetration ability**
 - X ray can penetrate wood, glass , paper, 1mm thickness Aluminium plate. But almost completely absorbed by a 1mm thick lead plate
 - Used lead sheets to protect from the X ray.

3. X ray travel with the speed of the light as that of other EM radiation in vacuum. ($c = 3.0 \times 10^8 \text{ ms}^{-1}$)
4. They cannot be reflected, refracted or deflected by magnetic or electric field as they do not possess any charge.
5. X ray cannot be focused on a single point like visible light.

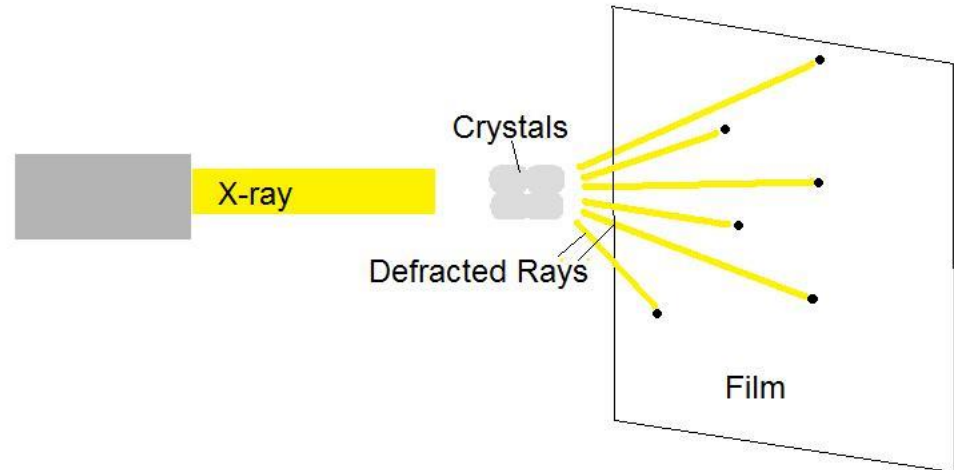
6. crystal lattice diffracts the X-ray beam differently.

- The wavelength of X-rays is on an atomic level and is much smaller than that of visible light.
- crystals could be used as a diffraction grating (the width should be as small as a few nanometers in length) for X-rays.
- crystal structure, in which the atoms are arranged in analogously, it acts like an X-ray diffraction grating.

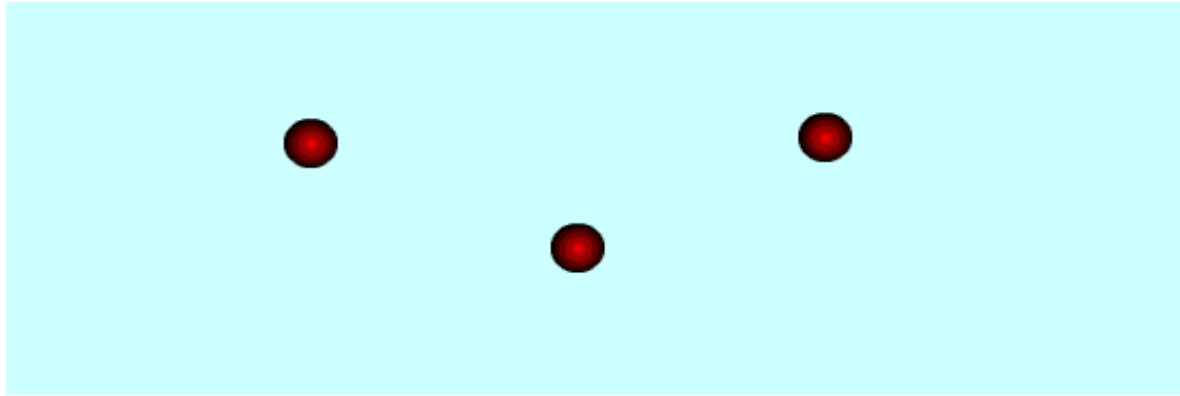
- X-ray diffraction is a very powerful tool used to study crystal structure. By examining the x-ray diffraction pattern, the type of crystal structure can be identified.



Diffraction Pattern



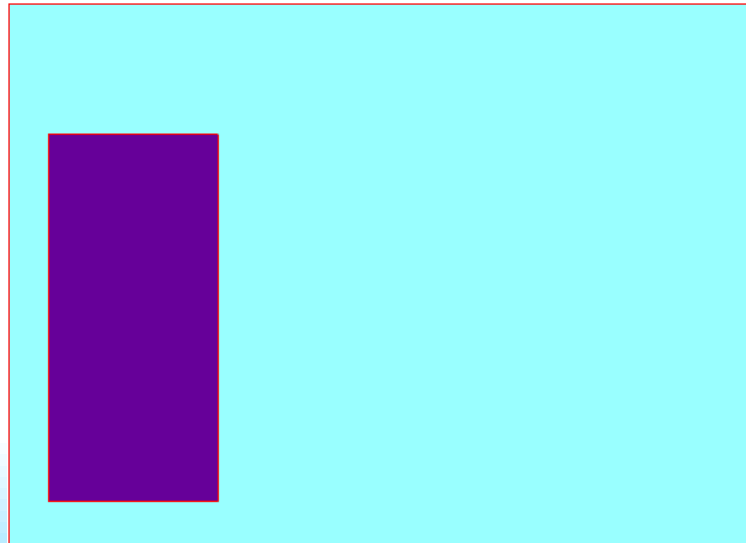
7. X ray photon (energy more than 34keV) interact with air and cause ionization in air



8. X ray are capable of producing an image on photographic film.



9. X rays fall upon certain materials visible light will be emitted called fluorescence.
10. Upon exposing a metallic surface to X ray radiation , due to the photo electric effect electrons are emitted.

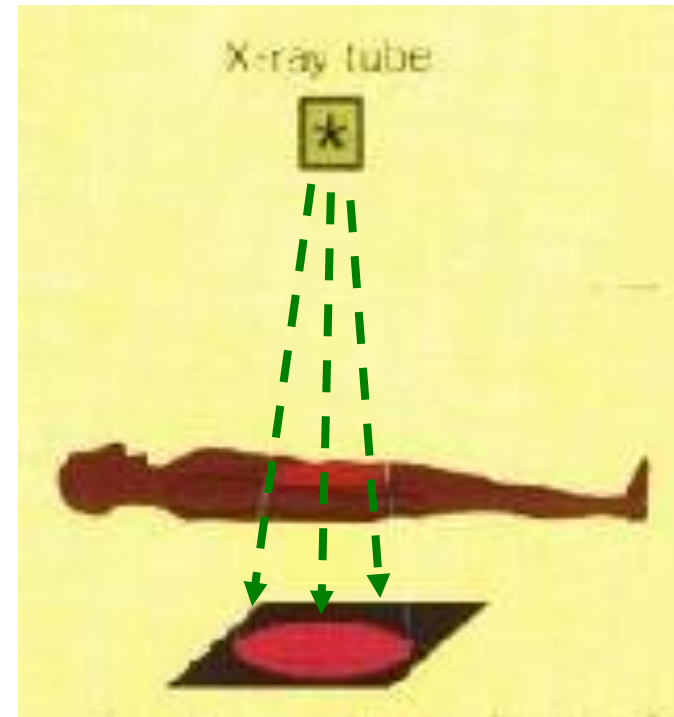
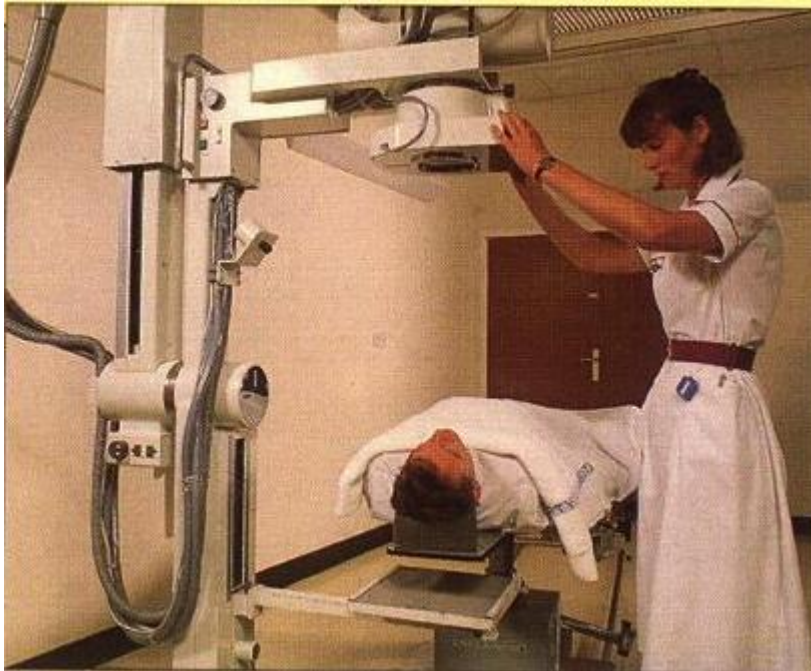


4. Applications

- a. Medical diagnosis
- b. X-ray use as a treatment
- c. Non destructive testing
- d. X-ray Fluorescence (XRF) Applications
- e. For Security Purposes

a. Medical diagnosis

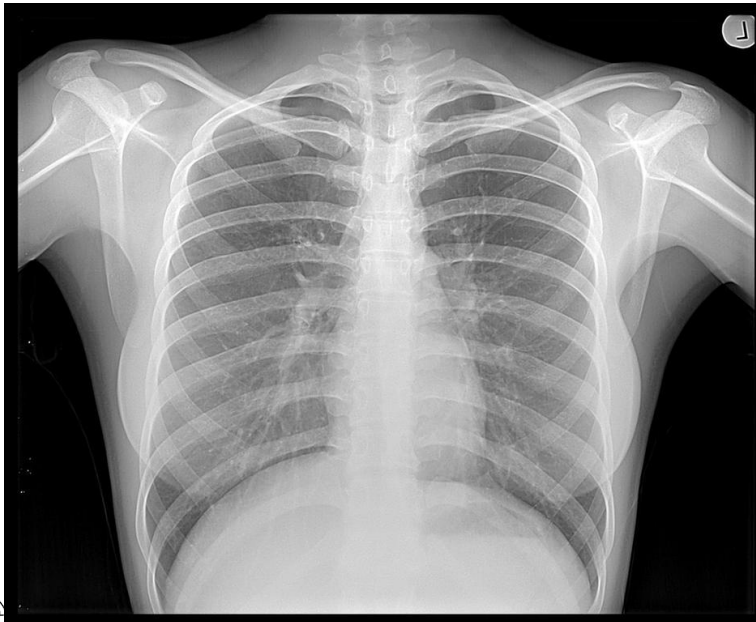
X-Ray Radiography- Medical imaging has led to improvements in the diagnosis and treatment of numerous medical conditions in children and adults.



Testing part shall be in-between the x ray source and x ray film.

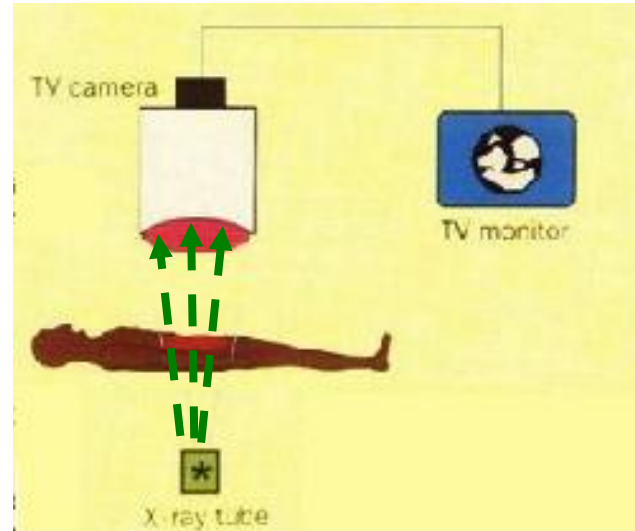
After the exposure, X ray film is processed using chemicals and obtained x ray radiograph.

(Mostly use for Radiography of Skull, Stomach, Spine, Hands, Legs, Teeth.)



a. Medical diagnosis

Fluoroscopy Testing



A continuous X-ray beam is passed through the body part and sent to a video monitor so that the body part and its motion can be seen in detail.

As the patient drinks the liquid barium (absorbed X ray radiation), the radiologist will watch the barium pass through the patient's digestive tract on a fluoroscope, a device that projects radiographic images in a movie-like sequence onto a monitor.



a. Medical diagnosis

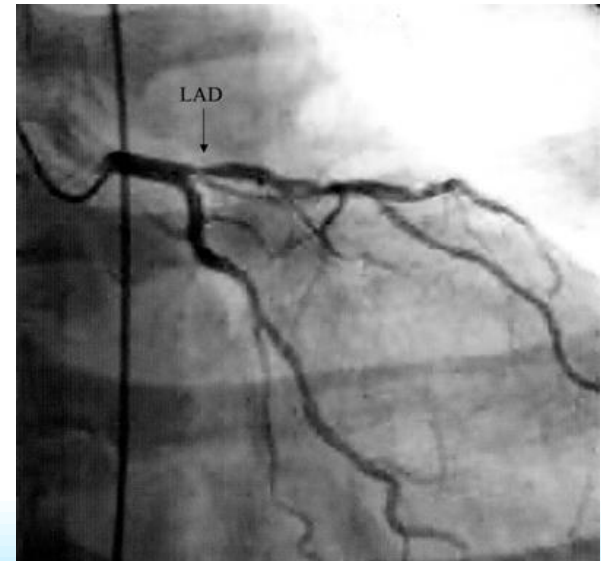
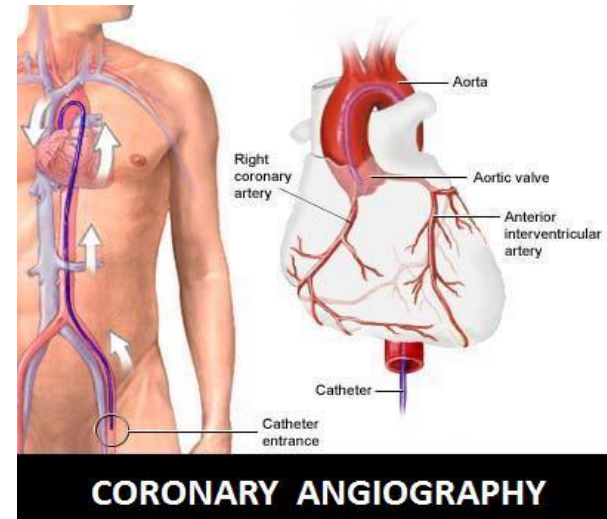
Angiography Testing

- An **angiogram** is an X-ray image of blood vessels after they are filled with a contrast material. An **angiogram** of the heart, a coronary **angiogram**, is the "gold standard" for the evaluation of coronary artery disease (CAD).
- A coronary **angiogram** can be used to identify the exact location and severity of CAD.



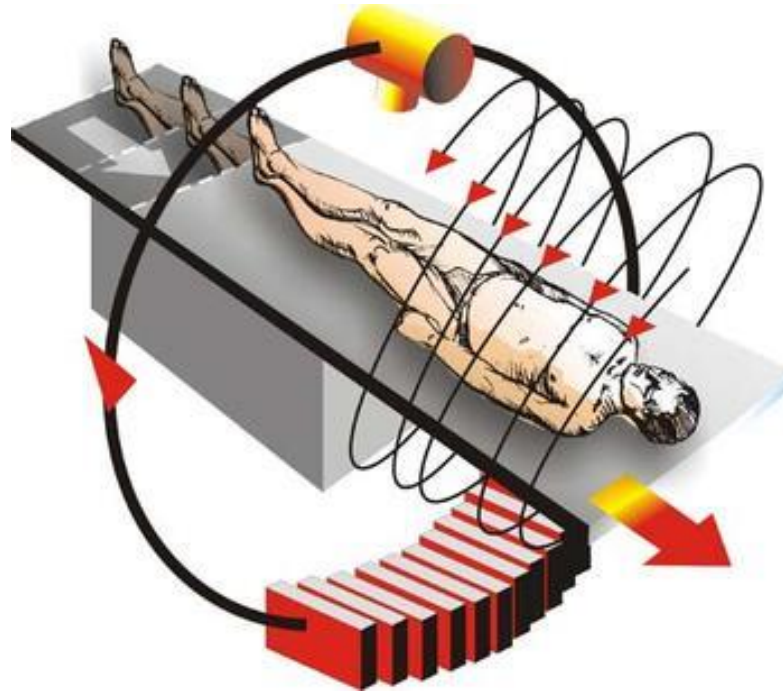
a. Medical diagnosis

Angiography Testing

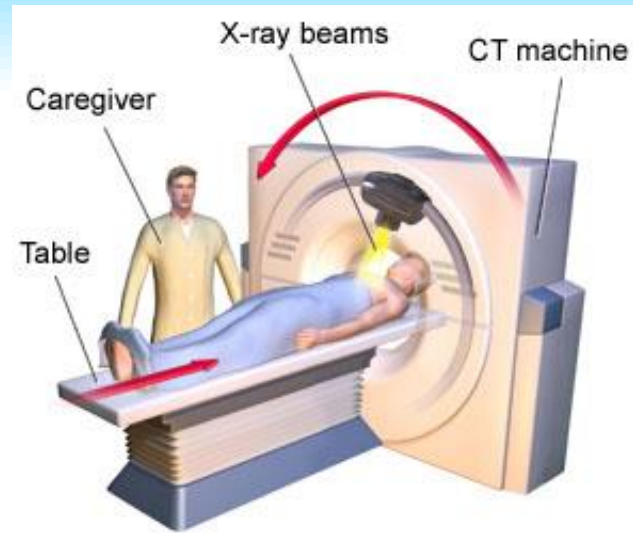


a. Medical diagnosis

Computed Tomography Scanning - (CT Scan)



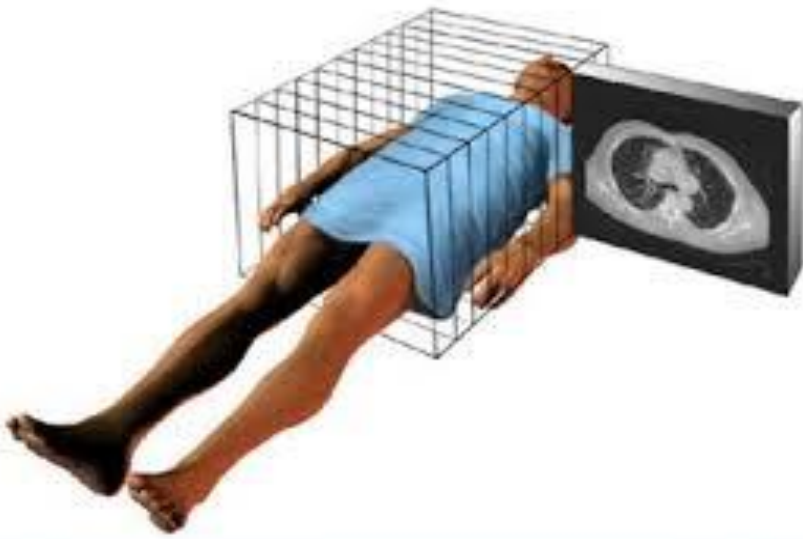
- The patient is inside the opening, an X-ray source and a detector assembly within the system rotate around the patient.



Computerized Axial Tomography Scan

- During rotation the X-ray source produces a narrow, fan-shaped beam of X-rays that passes through a section of the patient's body.
- The cross-sectional images generated during a CT scan can be reformatted in multiple planes.

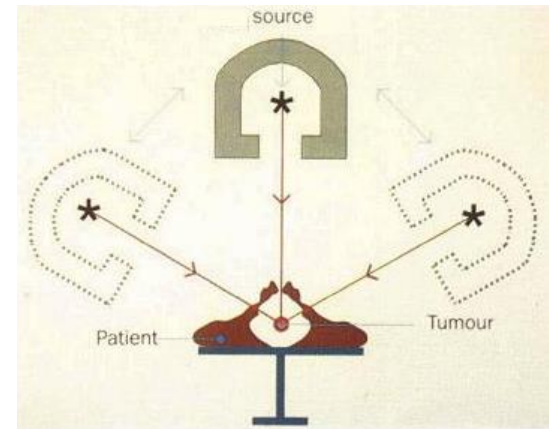
Computed Tomography image of the Body



- Advanced CT machines can generate three-dimensional images which can be viewed on a computer monitor.

b. X-ray use as a treatment

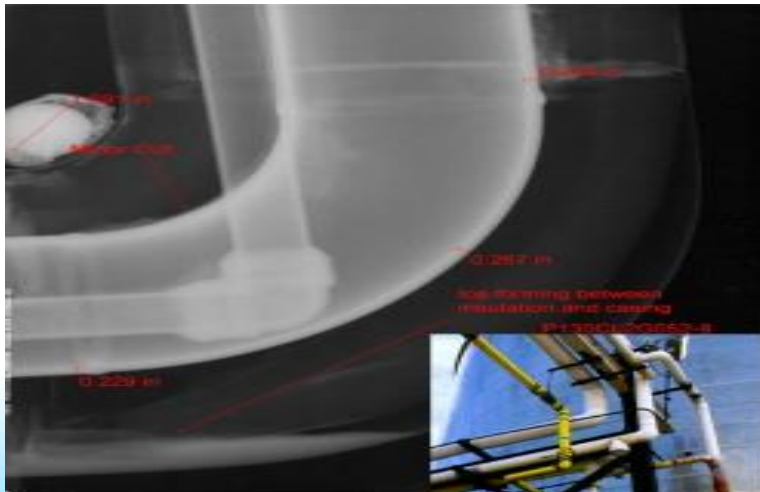
X-ray use as a treatment, called Radiotherapy



- Radiation **Therapy** to **Treat Cancer**
 - Radiation **therapy** (also called radiotherapy) is a **cancer treatment** that uses high doses of radiation to kill **cancer** cells and shrink tumors

c. Non destructive testing (NDT)

- **NDT** is a wide group of analysis techniques used in science and technology industry to evaluate the properties of a material, component or system **without causing damage**.
 - Nondestructive testing methods are routinely applied in industries where a failure of a component would cause significant hazard or economic loss, such as in transportation, pressure vessels, building structures, piping, and hoisting equipment.



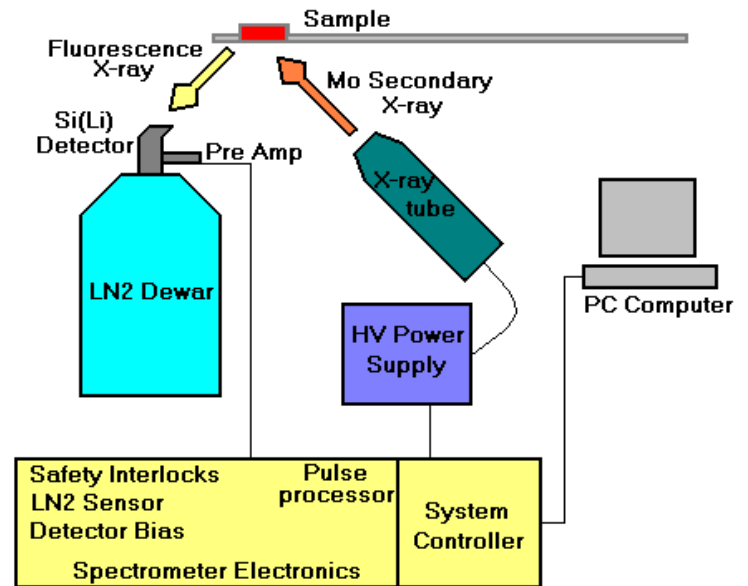
Several NDT techniques are used for this and **X-ray images in radiographic testing** are used to identify interior defects of components



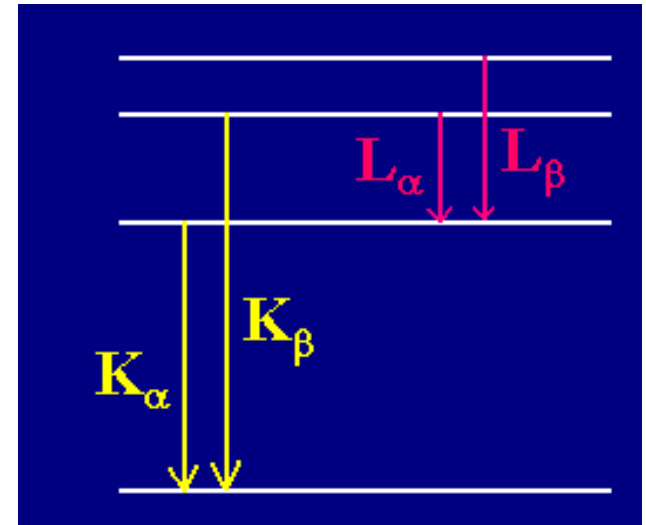
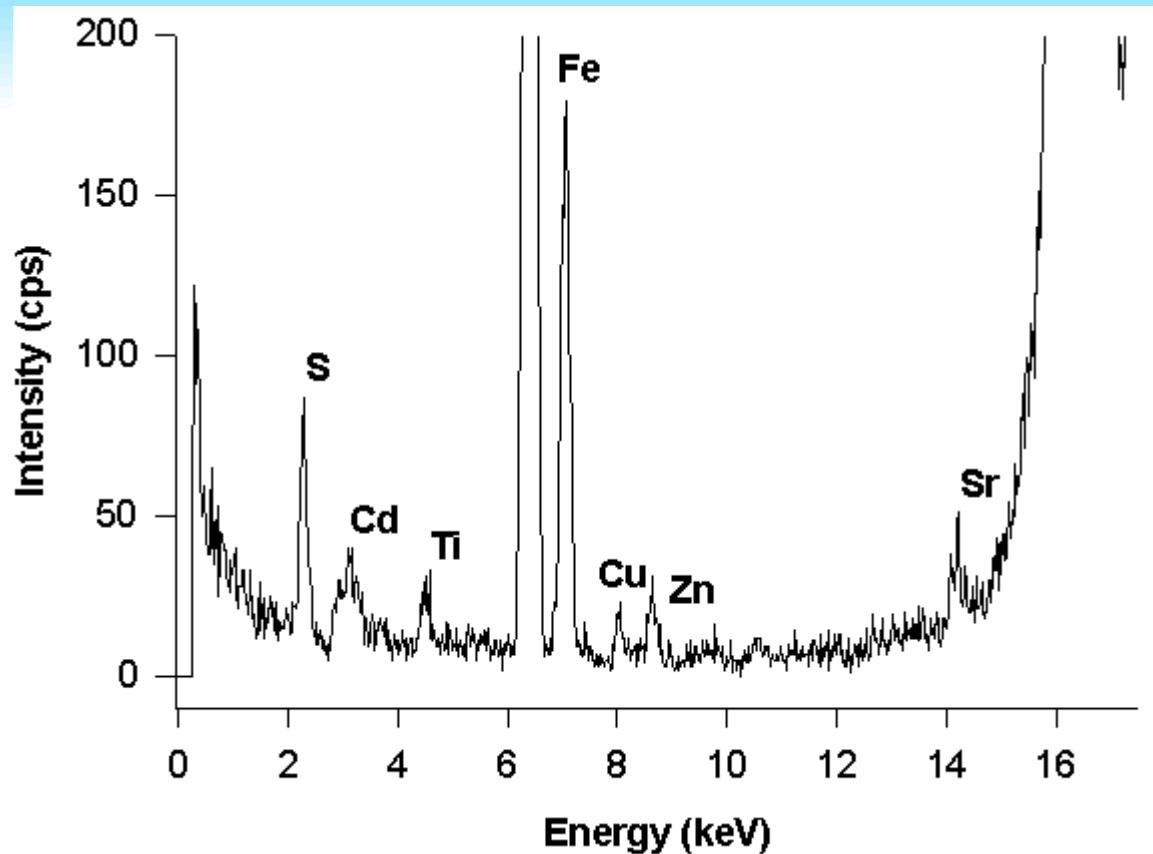
Practical situation of obtaining X- ray images

d. X-ray Fluorescence (XRF) Applications

- **XRF** technology is used to identify and analyze of elements in a wide variety of samples, qualitatively and quantitatively.
- In here, elements of the sample are identified by using emitted characteristic X-ray spectrum of each element due to incident X-rays on the sample.



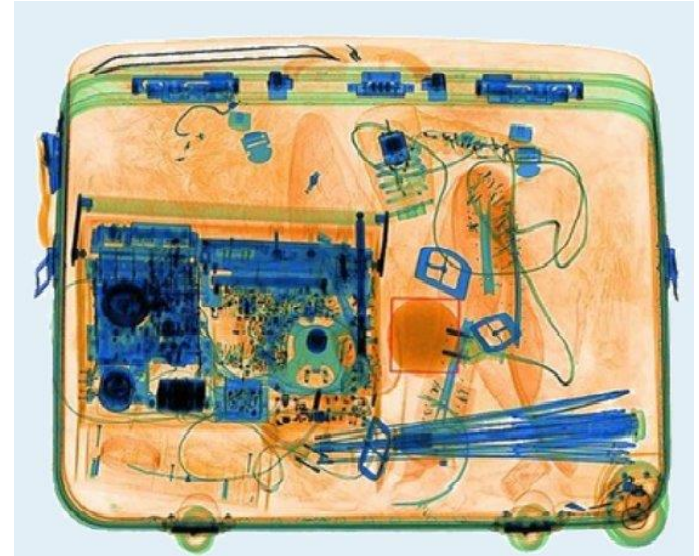
Typical Si(Li) Detector Instrument



Elements can be identified by using characteristic X-ray spectrum of XRF technology.

e. For Security Purposes

X-rays are commonly used in air ports and ports to identify things in the luggage without open them.



Thank You

